



University of Asia Pacific

Department of Computer Science and
Engineering **CSE 437: Robotics Lab**

LAB REPORT

Experiment Number: 03

Experiment Title: Obstacle Avoidance Maze-Solving Robot Using Arduino

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1. Experiment Name

Design and Implementation of an Obstacle Avoidance Maze-Solving Robot Using Arduino.

2. Objective

The main objective of this experiment was to design and implement an autonomous robot capable of avoiding obstacles and solving simple mazes without human assistance.

- To design and implement an autonomous obstacle avoidance robot.
- To interface ultrasonic sensors with **Arduino Uno**.
- To control DC motors using a motor driver module.
- To implement a basic maze-solving algorithm (Left-hand rule method).
- To enable the robot to navigate safely in an unknown environment without human intervention.

3. Apparatus / Hardware & Software Requirements

Hardware Components

- Arduino Uno (ATmega328P microcontroller)
- HC-SR04 Ultrasonic Sensor
- L293D Motor Driver Module
- DC Motors (2x)
- Servo Motor (for rotating ultrasonic sensor)
- Robot chassis with wheels
- Battery (7.4V–12V)
- Jumper wires
- Breadboard

Software Requirements

- Arduino IDE
- (Optional) Tinkercad / Wokwi for simulation

4. Circuit Diagram / Schematic

Motor Driver (L293D) → Arduino UNO

L293D Pin	Function	Arduino Pin
Pin 1	Enable Motor A	D5 (PWM)
Pin 2	Input 1	D8
Pin 7	Input 2	D9
Pin 9	Enable Motor B	D6 (PWM)

Pin 10	Input 3	D10
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Pin 15	Input 4	D11
Pin 16	Vcc1 (Logic 5V)	5V
Pin 8	Vcc2 (Motor Power)	External Battery +
Pin 4,5,12,13	GND	Arduino GND

Motor outputs:

- Pin 3 & 6 → Motor A
- Pin 11 & 14 → Motor B

Ultrasonic Sensor (HC-SR04)

Sensor Pin	Arduino Pin
VCC	5V
GND	GND
TRIG	D3
ECHO	D2

Servo Motor

Servo Wire	Arduino Pin
Signal (Orange/Yellow)	D4
VCC (Red)	5V
GND (Brown/Black)	GND

Power Connections

Component	Connection
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Arduino	USB power
Motor driver motor supply	External battery
All grounds	Common ground (important)

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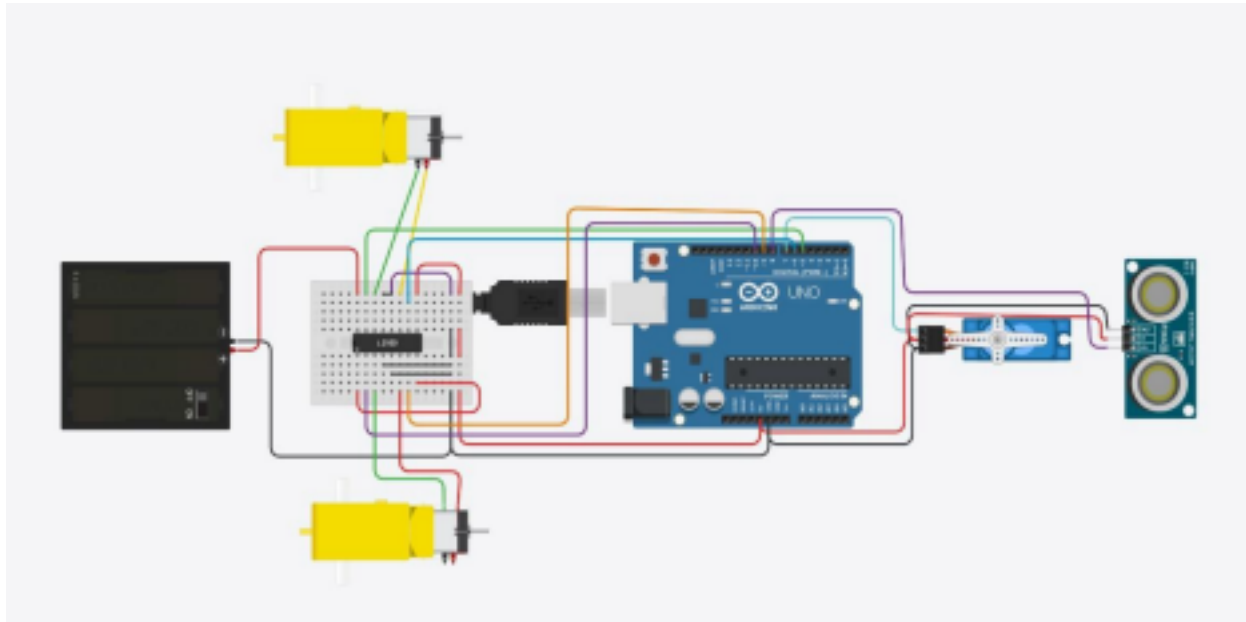


Fig 4.1: Circuit Diagram for obstacle avoidance maze-solving robot.

5. Code / Assembly Program

```
#include <Servo.h>
```

```
#define trigPin 9
```

```
#define echoPin 8
```

```
#define IN1 10
```

```
#define IN2 11
```

```
#define IN3 12
```

```
#define IN4 13
```

```
Servo myServo;
```

```
long duration;
```

```
int distance;
```

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```
int leftDistance;
```

```
int rightDistance;
```

```
void setup() {
```

```
    pinMode(trigPin, OUTPUT);
```

```
    pinMode(echoPin, INPUT);
```

```
    pinMode(IN1, OUTPUT);
```

```
    pinMode(IN2, OUTPUT);
```

```
    pinMode(IN3, OUTPUT);
```

```
    pinMode(IN4, OUTPUT);
```

```
    myServo.attach(6);
```

```
    myServo.write(90);
```

```
    Serial.begin(9600);
```

```
}
```

```
int getDistance() {
```

```
    digitalWrite(trigPin, LOW);
```

```
delayMicroseconds(2);  
  
digitalWrite(trigPin, HIGH);  
  
delayMicroseconds(10);  
  
digitalWrite(trigPin, LOW);  
  
  
duration = pulseIn(echoPin, HIGH);
```

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```
distance = duration * 0.034 / 2;  
return distance;  
  
}
```

```
void moveForward() {  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW); }
```

```
void moveBackward() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, HIGH);  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
}
```

```
void turnLeft() {
```

```
digitalWrite(IN1, LOW);  
digitalWrite(IN2, HIGH);  
digitalWrite(IN3, HIGH);  
digitalWrite(IN4, LOW); }
```

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```
void turnRight() {  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
}
```

```
void stopRobot() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, LOW);  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, LOW);  
}
```

```
void loop() {
```

```
    distance = getDistance();
```

```
    if (distance > 30) {
```

```
moveForward();
```

```
}
```

```
else {
```

```
stopRobot();
```

```
delay(300);
```

6

```
// Look Left
```

```
myServo.write(150);
```

```
delay(500);
```

```
leftDistance = getDistance();
```

```
// Look Right
```

```
myServo.write(30);
```

```
delay(500);
```

```
rightDistance = getDistance();
```

```
myServo.write(90);
```

```
// Left-hand rule for maze solving
```

```
if (leftDistance > rightDistance && leftDistance > 30) {
```

```
turnLeft();
```

```
delay(500);
```

```
}
```

```
else if (rightDistance > 30) {
```



```

turnRight();
delay(500);
}
else {
moveBackward();
delay(500);
}

}

}

```

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6. Output / Observations

During the experiment, the following behaviors were observed:

- When powered ON, the robot moves forward.
- If an obstacle is detected within 30 cm:
 - Robot stops immediately.
 - Servo rotates ultrasonic sensor left and right.
 - Measures distance in both directions.
 - Chooses the clearer path (Left-hand priority rule).
- If both sides are blocked, robot moves backward and re-evaluates.
- The robot successfully navigates simple maze paths.

7. Result

The obstacle avoidance maze-solving robot was successfully designed and implemented using Arduino Uno and the HC-SR04 ultrasonic sensor.

The robot was able to:

- Detect obstacles accurately.
- Change direction autonomously.
- Solve simple maze paths using a left-hand rule algorithm.

Hence, the objective of the experiment was achieved successfully.

8. Conclusion

Through this experiment, we gained practical knowledge about:

- Interfacing ultrasonic sensors with microcontrollers.
- Controlling DC motors using the **L293D** motor driver.
- Implementing decision-making algorithms in embedded systems.
- Basics of autonomous robotic navigation.

This project enhanced practical knowledge in embedded systems, robotics, and sensor-based automation.